

Project Frontier: Feasibility of Fabrik, a 3D-DRAM Decode Accelerator

A full-stack simulation study on the ATLAS (MICRO '26) infrastructure.

Evidence tags used throughout: [measured] = run in this project on real tools (Ramulator/BookSim/HotSpot cycle or thermal simulation, or public API measurements by named third parties); [cycle] = ATLAS cycle-level simulation; [analytical] = this project's validated analytical model; [external] = sourced public figure; [estimated] = derived from external figures with stated method; [assumed] = modeling assumption, carried through the uncertainty analysis (§20).

Abstract

We ask whether a purpose-built decode accelerator ("Fabrik") — small compute die stacked on high-bandwidth 3D DRAM — can serve modern long-context MoE models at Cerebras-class interactivity (1,000–2,000 output tokens/s/user) at batch 1–4, with NVIDIA-class or better cost per token and materially better energy per token. We build an independent analytical simulator validated to within $\pm 20\%$ against ATLAS cycle-level simulation [cycle], calibrate its memory model with Ramulator measurements per access pattern [measured], and sweep 64,800 design points across bandwidth (12–150 TB/s), compute (0.25–4 PFLOP/s), capacity (128–512 GB), and SRAM (8–256 MB) for DeepSeek V4-Flash (284B/13B-active, hybrid compressed-sparse attention) and MiniMax M3 (428B/23B-active, MiniMax Sparse Attention), both at 1M context.

Finding: SUPPORTED for DeepSeek-V4-Flash-class workloads; PLAUSIBLE BUT UNPROVEN for MiniMax-M3-class workloads. A 50 TB/s / 1 PFLOP/s / 256 GB Fabrik serves V4-Flash at 2,560 TPS/user (B=1, 128K ctx) in ~480 W, ~5.3 tok/J, and \$0.16/1M output tokens at \$25K system cost — a (interactivity, cost) point roughly 8–20× faster than measured GPU serving of the same model at comparable or lower cost/token [analytical], [external]. M3 reaches 2,000 TPS/user only at ~100 TB/s and ~1 kW [analytical], brushing our falsification thresholds; its 428 GB fp8 weight footprint also makes single-device capacity the binding constraint.

1. Motivation

Autoregressive decode of MoE LLMs at batch 1–4 is memory-bandwidth-bound: every generated token must stream the activated weights and the attention working set. GPUs amortize weight traffic across large batches, sacrificing per-user speed; wafer-scale SRAM machines buy speed with silicon area. A logic-on-3D-DRAM part occupies a third position: DRAM-class capacity and cost per bit with SRAM-class bandwidth. The commercial hypothesis is that decode — disaggregated from GPU prefill — is the workload where that position wins.

2. Research question

| *Can a purpose-built 3D-DRAM decode accelerator achieve ~1,000–2,000 output*

tokens/s/user at NVIDIA-class or better cost/token and materially better

energy efficiency for modern large MoE models at batch sizes 1–4?

3. Workloads

Both workloads are reconstructed **exactly** from official sources (archived with revisions in `models/source_of_truth/`; per-parameter provenance in `models/model_manifest.json`).

3.1 DeepSeek V4-Flash `[external: official config + inference/model.py, rev 60d8d70]`

43 layers, hidden 4096, all layers MoE (256 routed FP4 experts, top-6, 1 shared FP8, inter 2048; first 3 layers hash-routed). Attention is single-shared-KV MQA with head_dim 512 (rope 64), in three per-layer modes set by `compress_ratios`: 2 sliding-window(128) layers; 21 **CSA** layers (4:1 learned-compressed KV + a 64-head×128-dim FP4 **Lightning-Indexer** that scans *all* t/4 compressed entries and selects top-512); 20 **HCA** layers (128:1 compression, attends *all* t/128 entries). Hyper-connections (hc_mult 4) widen the residual stream 4×. Our reconstruction totals 284.3B params (official: 284B) and ~13.6B active (official: 13B) `[analytical, validated]`.

Decode traffic at B=1 `[analytical, validated against closed forms]`: **11.2 GB/token** at 32K, of which routed experts are only 3.4 GB — the *dense* attention-side projections (wq_b/wo_a/wo_b at 64×512 head geometry, 4.6B params) are the largest stream. KV is nearly flat with context (compressed); the **indexer scan grows linearly** and reaches 0.37 GB/token at 1M — V4-Flash's context-scaling bottleneck is index-scan bandwidth, not KV. Total KV+index state: **3.6 GB/user at 1M** — trivially small.

3.2 MiniMax M3 `[external: official config rev f0e1c1e; MSA mechanics from arXiv:2606.13392]`

60 layers, hidden 6144, GQA 64Q/4KV heads ×128. Layers 0–2: dense FFN (12288) + **full attention**; layers 3–59: MoE (128 experts, top-4, 1 shared, inter 3072) + **MSA** (top-16 blocks of 128 tokens + local block, 4 index heads ×128, max-pool block scoring, KV-outer-gather-Q). Reconstruction: 426B (official 428B; delta = the 7 MTP modules we model off). The checkpoint is bf16; serving precision is unpublished — we model `exact_fp8` (base, `[assumed]`) and `exact_bf16` variants.

Decode traffic at B=1, fp8: **26 GB/token** at 32K (routed+shared experts 12.9 GB, attention/dense projections 13 GB), rising to 29.4 GB at 1M — of which **3.1 GB/token is the three full-attention layers alone**. Per-user KV state: **64.5 GB (fp8) / 129 GB (bf16) at 1M** — capacity, not bandwidth, is M3's binding constraint on a single device. In bf16, M3 does not fit any swept capacity (needs 856 GB weights) — every bf16 point is infeasible `[analytical]`.

4. Decode characterization

Bottleneck regimes (B=1) `[analytical]`:

workload	32K	128K	1M
V4-Flash	weight-bound (dense projections > experts)	weight-bound	weight-bound, index-scan emerging (24% of added traffic; compute of attention+indexer rises 4.4×)

M3 (fp8)	weight-bound (experts+dense)	weight-bound	weight-bound with KV-bound tail (full-attn layers = 3.1 GB/token)
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Arithmetic intensity stays at 2.2–12 FLOP/byte — an order below GPU-optimal — confirming decode at $B \leq 4$ belongs on a bandwidth-first machine.

5. Analytical methodology

Per-operator $T = \max(\text{bytes}/(\text{BW} \cdot \eta(\text{pattern})), \text{FLOPs}/(\text{peak} \cdot \text{util}(\text{org}, B))) + \text{fixed}$, dependency-serialized with double-buffered overlap inside each operator; a global $\max(\Sigma T_{\text{mem}}, \Sigma T_{\text{comp}})$ lower bound brackets pipelining uncertainty. Batch composition: shared weight streams amortize; per-user KV/index/attention terms scale with B ; routed-expert streams follow the expert-reuse model. Millisecond evaluation $\rightarrow$ 64,800-point sweeps in ~ 7 s.

Small-batch MoE reuse (Part VII): with E experts and top- k routing, $E[\text{unique}] = \Sigma_e 1 - (1 - p_e)^B$. At $B=4$: V4-Flash 23.2 unique of 24 naive (3% saving), M3 15.3/16 (4%) [analytical]; even Zipf- $\alpha=1$ hot-expert routing saves only $\sim 21\%$ [analytical, synthetic traces – real routing traces are not public, labeled] . **Expert weights effectively do not amortize at $B \leq 4$** — the premise of a bandwidth-first design holds.

6. ATLAS methodology

ATLAS (MICRO '26) provides the cycle-level substrate: per-core compute/SRAM/NoC models + Ramulator2 with a custom HBDRAM 3D-DRAM device + BookSim + HotSpot thermal (docs/ATLAS_ARCHITECTURE.md). We verified (simulator/src/core/core.cpp:156) that **each core owns a private DRAM vault** — the stock cloud chip is $16 \text{ cores} \times (16\text{ch} \times 1024 \text{ pin @ } 500 \text{ Mbps} = 1.024 \text{ TB/s}) = 16.4 \text{ TB/s}$, 128 GB. Heavy runs execute on 32-core cloud workers (Modal) from an image of this patched tree; upstream kick-the-tires and one DRAM-DSE family reproduce cleanly (BASELINE.md) [cycle] .

Validation (fig13, results/processed/validation_summary.csv): the analytical model, with zero tuned parameters, reproduces ATLAS cycle-level end-to-end decode latency within **$0.82\times$ (opt-66B dense, TP8, B=8)** and **$1.16\times$ (Mixtral-8 $\times$ 22B MoE, TP8/EP8, B=8)** on the stock chip [cycle] .

Memory-pattern calibration [measured] (calibrate_mem_eff.py, Ramulator test_dram on the 1 TB/s vault): well-tiled sequential weight streams achieve **0.80** of peak; MSA-style 128-token block gathers **0.77**; V4-style scattered 576 B row gathers **0.29**; index/HCA scans **0.81**. Naive tiling collapses streams to 0.20–0.31 — the SRAM-sensitivity model is anchored to this measured spread. Burst dependency latencies: 512-row CSA gather 7.7 μs , MSA 17-block gather 2.7 μs .

7. Fabrik architecture

A Fabrik point is (BW, compute, capacity, SRAM, compute-org), physically realized as N vault-cores (per-core 1.024 TB/s ATLAS-validated vault; $N = \text{BW}/1.024$), per-core MAC count set by the compute target, per-core stack density by capacity (feasible 1–64 GB/core). The flagship configurations:

	V4-Flash flagship	M3 flagship
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DRAM BW	50 TB/s (49 vaults)	100 TB/s (98 vaults)
Compute (FP8)	1 PFLOP/s	2 PFLOP/s
Capacity	256 GB	512 GB
SRAM	64 MB	128 MB
org	hybrid (reconfig systolic + GEMV lanes + index engine)	hybrid

8. 3D-DRAM architecture

Not all bytes are equal (§6 calibration). The DSE therefore prices each operator's traffic at its measured pattern efficiency. The **Raptor calibration case** (`calibration/raptor/`) checks realism at scale: 104 vaults → 106.5 TB/s peak reproduces a documented logic-on-DRAM card class (~105 TB/s); sustained streaming 86 TB/s (81%); ns-scale per-flit streaming latency, consistent with the ~2.5 ns reference figure [measured] . All parameter divergences from the real chip are enumerated in the module header — this is a behavioral sanity check, not a product reproduction.

9. Small-batch compute architecture

[analytical; org utilization parameters assumed, anchored to ATLAS area/power scalars and swept $\pm 30\%$ in MC] At B=1–4 the workload is GEMV-shaped. Fixed 128×128 systolic arrays waste rows (util ~12.5% at M=1); reconfigurable aspect ratios recover to ~60–80%; dedicated GEMV lanes ~90%. Speedup of hybrid vs the ATLAS baseline engine at B=1, 1M ctx: **1.57× for V4-Flash** (the index engine is the difference — see §10) and **1.05× for M3**. Compute utilization at the flagship points stays below 15% of peak FP8 — decode compute exists to keep up with the stream, not to saturate.

10. Sparse-attention acceleration

V4-Flash: the indexer scan (all t/4 entries × 64 heads × 128 dims, FP4) is 86 GFLOP + 0.37 GB per token at 1M and runs poorly on generic engines (index_util 0.3): a dedicated scan/top-k engine (util 0.9) is the main term in the 1.57× hybrid speedup — **a specialized indexing unit is worthwhile for V4-class CSA**. M3: MSA's block-max scoring is small (n_blocks×4×128) and its gathers are 128 KB-contiguous (0.77 efficiency [measured]) — **specialized hardware buys little (~5%)**; commodity engines suffice.

11. Power and thermal

Power model (per-subsystem pJ/bit + org W/TFLOP; constants [assumed] , MC-swept): at the V4 flagship point, DRAM array reads dominate (~55% of energy), then interface, compute, SRAM, NoC — "memory-array energy cannot be omitted" is quantitatively confirmed. HotSpot thermal [measured] (PyTA, liquid cooling, 8 memory layers, 85 °C threshold): the 40–50 TB/s flagship class at its natural ~480–560 W runs 84–92 °C — thermally valid at 1 GHz or with one 100 MHz derate step at a 400 W envelope. Full envelope×design table:

`results/processed/thermal_summary.csv` .

12. Prefill/decode disaggregation

$T_{\text{request}} = T_{\text{prefill}}(\text{GPU}) + T_{\text{handoff}} + N_{\text{out}} \times T_{\text{decode}}$ with GPU prefill 55k tok/s [estimated, labeled]. V4-Flash handoff is trivial: 3.6 GB KV at 1M moves in 14 ms at 256 GB/s (<0.1% of a request). M3 is the stress case: 64 GB KV at 1M takes 1.0 s on a 64 GB/s link (5% of a short-output request, TTFT +1 s) — M3 disaggregation wants ≥ 256 GB/s links or KV-resident session affinity. Link latency (100 ns–10 μ s) is irrelevant at these sizes [analytical].

13. Experimental setup

64,800-point analytical DSE (10 BW $\times$ 8 compute $\times$ 5 capacity $\times$ 6 SRAM $\times$ 2–3 model modes $\times$ 3 contexts $\times$ 3 batches), calibrated as §6; representative-point cycle/ thermal validation; serving simulation (continuous batching, cap 4); TCO per Part XXIII formulas. All reproducible: `experiments/run_dse.py`, `run_pareto.py`, `run_min_hw.py`, `run_thermal.py`, `run_serving.py`, `run_orgs.py`, `run_disagg.py`, `run_uncertainty.py`, `make_figures.py`.

14. Performance results

Flagship points [analytical]:

config	ctx	B	decode step	TPS/user	aggregate
V4 flagship (50 TB/s)	128K	1	0.39 ms	2,562	2,562
V4 flagship	128K	4	0.67 ms	1,484	5,937
V4 flagship	1M	1	0.46 ms	2,184	2,184
M3 flagship (100 TB/s)	128K	1	0.43 ms	2,310	2,310
M3 flagship	128K	4	0.92 ms	1,088	4,352
M3 flagship	1M	1	0.49 ms	2,040	2,040

Minimum hardware for targets (B=1/B=4 @128K) [analytical]: V4-Flash 1,000 TPS: **24/32 TB/s**; 2,000 TPS: **40/75 TB/s**. M3-fp8 1,000 TPS: **40/100 TB/s**; 2,000 TPS: **100 TB/s / not reachable at B=4 within the swept space**. Serving simulation: at ~50–70% device occupancy the V4 flagship holds p50 $\approx$ 2,200 TPS/user with p95 TPOT < 0.7 ms; saturation degrades to the B=4 floor (~1,480) with queueing, not TPOT, absorbing load.

15. Energy results

V4 flagship: 5.3 tok/J at B=1, 10.7 tok/J at B=4 (128K); M3 flagship 2.3 / 3.7 tok/J. GPU reference: an H200/B200-class node serving V4-Flash at measured provider speeds implies ~0.5–1 tok/J [estimated from external throughput and node power, labeled] — Fabrik's advantage is roughly **5–10 $\times$ at iso-model** at B=4. Subsystem split at flagship: DRAM array 55%, interface 9%, compute 7%, SRAM 6%, static 9%, rest NoC/controller. Reducing bandwidth below the interactivity-minimum always improves tok/J slightly but costs TPS/user; the energy-optimal point sits just above the target-TPS bandwidth minimum (§17, fig04).

16. Cost / TCO results

At \$25K system cost, 4-year life, PUE 1.25, \$0.08/kWh, 60% utilization [assumed, swept]: V4 flagship **\$0.165/1M output tokens at B=1** and **\$0.072/1M at B=4**; M3 flagship \$0.19/\$0.10. Measured GPU serving of the same model spans \$0.168–0.66/1M [external]. NRE: even \$400M amortizes below GPU-reference cost/token by ~10k units at B=4 economics; \$50M by ~1k units [analytical].

17. Design-space exploration

fig02: at B=1 the TPS contours are nearly vertical — **bandwidth is the only first-order axis**; ≥ 0.5 PFLOP/s of compute suffices until 1M-context attention/indexer FLOPs matter. At B=4 contours bend: 2,000 TPS needs both (75 TB/s and ~1 PFLOP/s for V4). Extra bandwidth beyond the target minimum buys headroom linearly until compute-bound onset (V4 B=4: ~125 TB/s at 2 PFLOP/s). 100 TB/s is *not* overprovisioned for M3 (it is the 2,000-TPS minimum) but is $\sim 2.5\times$ overprovisioned for V4-Flash at B=1.

18. Pareto frontier

3,481 non-dominated points (results/pareto/). The 2,000-TPS frontier for V4 is anchored by 40–75 TB/s / 0.25–1 PFLOP/s / 32–64 MB designs at 380–750 W — i.e., the flagship class, not the 150 TB/s corner. SRAM beyond 64 MB buys <2% at 50 TB/s (fig10); capacity is pure feasibility (V4: 128 GB suffices to B=4 @1M; M3: 512 GB admits only B=1 @1M).

19. Comparison with existing accelerators

baselines/external.csv (all sourced; comparison classes labeled). Same-model GPU serving: 107–306 TPS/user, \$0.168–0.66/1M. Cerebras: 1,800–4,400 TPS/user on *different* (dense/smaller) models [external, market reference] at \$0.85–1.20/1M-class prices; no public Cerebras/Groq offering of V4-Flash/M3 exists, so same-model comparison with wafer-scale is not possible — the conclusion rests on the same-model GPU comparison plus Fabrik's absolute numbers, and survives this gap.

20. Sensitivity analysis

2,000-sample Monte Carlo on the V4 flagship (10 variables, triangular low/base/high): TPS/user p5–p95 = **2,103–2,689** (target 2,000 holds at p5); tok/J 3.4–5.8; \$/1M 0.14–0.35; power 400–739 W. Top-5 drivers (by \$/1M swing): hardware cost, fleet utilization, sequential-stream efficiency, DRAM array pJ/bit, compute utilization scale. No sampled combination pushed the V4 flagship below 1,900 TPS/user.

21. Limitations

1. V4/M3 cycle-level validation is by proxy: the analytical model is

cycle-validated on ATLAS-native models ($\pm 20\%$) and pattern-calibrated on Ramulator; the exact V4/M3 operator DAGs run analytically only.

2. Compute-org utilizations and all power constants are assumptions (swept);

no RTL exists.

3. ATLAS itself ships anonymized parameters (its AE docs); absolute cycle

numbers are trend-faithful, not silicon casting.

4. Routing traces are synthetic; real V4/M3 expert distributions unpublished.

5. M3 serving precision (fp8) is our assumption; bf16 makes M3 infeasible on a single device of any swept capacity.

6. GPU tok/J and prefill throughput references are estimates from public data.

7. Multi-device Fabrik (which would relieve M3 capacity) is future work.

22. Conclusions — answers to the mandated questions

Architecture. (1) Min BW for 1,000 TPS/user: 24 TB/s (V4, B=1) / 40 TB/s (M3-fp8, B=1). (2) For 2,000: 40 / 100 TB/s. (3) ~100 TB/s is required for M3-2000 but ~2.5× overprovisioned for V4 B=1. (4) Bandwidth stops helping at the compute-bound knee (V4 B=4: ~125 TB/s @2 PFLOP/s) — and earlier, at the target, for energy. (5) 0.25–0.5 PFLOP/s suffices at B=1; ~1–2 PFLOP/s at B=4/1M. (6) Compute-bound only at B=4 + long context + low compute. (7) 32–64 MB SRAM; beyond that <2%. (8) Yes for B=1–4: hybrid GEMV/reconfig gives 1.45–1.57× over a fixed engine (V4). (9) V4: yes (index engine is most of the win); M3: no (~5%).

Workload. (10) V4-Flash: weight-bound at 32K/128K; index-scan becomes the growth term at 1M (KV stays negligible). (11) M3: weight-bound at 32K/128K; at 1M the 3 full-attention layers make it KV-heavy (3.1 GB/token) and capacity-bound (64 GB/user). (12) **V4-Flash is the better Fabrik match** — 2.3× less traffic/token, 18× less KV state, fits 256 GB.

Interactivity. (13–16) V4: 1,000 TPS at B=1 ✓ (24 TB/s), B=2 ✓, B=4 ✓ (32 TB/s); 2,000 ✓ (40–75 TB/s). M3-fp8: 1,000 ✓ (40–100 TB/s); 2,000 only at B=1 with 100 TB/s. (17) B=1 → 4 degradation: V4 –42% (2,562 → 1,484); M3 –53% (2,310 → 1,088).

Power. (18) V4: yes — 2,000 TPS/user at 377–750 W. M3: 2,000 TPS needs ~900–1,000 W: *outside* 500–800 W. (19) 5.3–10.7 tok/J (V4), 2.0–3.7 (M3). (20) DRAM array (~55%). (21) Yes: past the target-TPS minimum, more bandwidth costs tok/J for nothing the SLA needs.

Economics. (22) ≤\$40K matches the cheapest same-model GPU route at B=1; at B=4, even \$100K does. (23) \$0.072–0.165/1M (V4 flagship, 60% util). (24) Batching 1 → 4 cuts \$/token 2.3× while holding 1,484 TPS/user. (25) \$50M NRE amortizes at ~1k units; \$400M by ~10k units.

(26) Thesis verdict:

| *A memory-centric accelerator can provide Cerebras-class interactivity with

| NVIDIA-class or better economics and materially better energy efficiency

| for B=1–4 decode.*

SUPPORTED for DeepSeek-V4-Flash-class workloads (compressed-attention, low-active-parameter MoE): every element of the claim holds simultaneously at the 50 TB/s flagship point, with p5 Monte-Carlo margin above the 2,000-TPS target and cost/token at-or-below the best same-model GPU route. **PLAUSIBLE BUT**

UNPROVEN for MiniMax-M3-class workloads: interactivity and economics hold at B=1 with fp8 weights, but the 2,000-TPS point needs ~1 kW (falsification criterion grazed), B=4 @1M violates single-device capacity, and the fp8-serving assumption is unverified. The falsification checklist (Part XXXIII) fires on no V4 criterion and on two M3 criteria (power; capacity-dominated economics at 1M).

Falsification checklist (Part XXXIII)

crit ^{erion}	V4-Flash	M3-fp8
<25% usable BW at B=1–4	no (51–63% modeled, patterns measured 29–81%)	no (60–70%)
>4–5 PFLOP/s needed to consume BW	no ($\leq 1\text{--}2$)	no (≤ 2)
>1 kW for target TPS	no (≤ 750 W)	yes at 2,000 TPS (~1 kW)
large B=4 TPS/user degradation	–42% (acceptable, still >1,000)	–53% (1,088 at 128K; fails @1M capacity)
capacity/package dominates TCO	no	partially (512 GB, B=1-only @1M)
communication dominates sub-ms decode	no ($< 0.1\%$)	no at ≥ 256 GB/s links
specialized compute buys little	no ($1.57\times$)	yes ($1.05\times$) — commodity engines fine for M3
\$/token worse than GPU serving	no (at/below best route)	no (B=1 parity, B=4 better)